

Topological Quantum Computation with Anyons

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Review

Fusion and Braiding

Examples

Quantum Computation with Anyons

Summary

Quantum logic gates

$$\begin{aligned}
 |\psi\rangle &= v_0 |0\rangle + v_1 |1\rangle, \\
 |\psi\rangle &= v_{00} |00\rangle + v_{01} |01\rangle + v_{10} |10\rangle + v_{11} |11\rangle.
 \end{aligned}
 \tag{1}$$

Pauli-X (X)			$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
Pauli-Y (Y)			$\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$
Pauli-Z (Z)			$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$
Hadamard (H)			$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$
Phase (S, P)			$\begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$
$\pi/8$ (T)			$\begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}$
Controlled Not (CNOT, CX)			$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$

Definition of the 2d toric code model

For each vertex v and plaquette p we define a vertex operator $A_v = \prod_i \sigma_x^i$ and a plaquette $B_p := \prod_j \sigma_z^j$ acting on adjacent edges. Here $\sigma_x^i = \cdots \otimes 1 \otimes \sigma_x \otimes 1 \otimes \cdots$ is the operator that acts on $\mathcal{H}_i$ as σ_x and acts on other local Hilbert spaces as identities. For example, the operators in following figure are

$$A_v = \sigma_x^1 \sigma_x^2 \sigma_x^3 \sigma_x^4, \quad B_p = \sigma_z^3 \sigma_z^4 \sigma_z^5 \sigma_z^6.$$

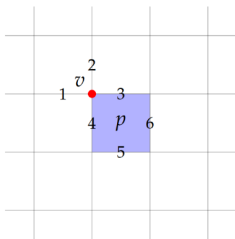


Figure: The 2d toric model.

Toric code anyons

- For every vertex v_0 and plaquette p_0 there is a state $|\psi_{v_0,p_0}\rangle$ satisfying the following property:

$$\left\{ \begin{array}{l} A_{v_0} |\psi_{v_0,p_0}\rangle = -|\psi_{v_0,p_0}\rangle, \\ A_v |\psi_{v_0,p_0}\rangle = |\psi_{v_0,p_0}\rangle \text{ for all vertices } v \neq v_0, \\ B_{p_0} |\psi_{v_0,p_0}\rangle = -|\psi_{v_0,p_0}\rangle, \\ B_p |\psi_{v_0,p_0}\rangle = |\psi_{v_0,p_0}\rangle \text{ for all plaquettes } p \neq p_0. \end{array} \right.$$

Then all states $|\psi_{v_0,p_0}\rangle$ generate the same topological excitation, denoted by f .

- We have found 4 different topological excitations: $\mathbb{I}$, e , m , f .
- Intuitively, if an e particle and an m particle are adjacent, they can be viewed as a single topological excitation f .

Toric code anyons

- The definition of the excitation f :

$$\begin{cases} A_{v_0} |\psi_{v_0, p_0}\rangle = -|\psi_{v_0, p_0}\rangle, \\ A_v |\psi_{v_0, p_0}\rangle = |\psi_{v_0, p_0}\rangle \text{ for all vertices } v \neq v_0, \\ B_{p_0} |\psi_{v_0, p_0}\rangle = -|\psi_{v_0, p_0}\rangle, \\ B_p |\psi_{v_0, p_0}\rangle = |\psi_{v_0, p_0}\rangle \text{ for all plaquettes } p \neq p_0. \end{cases}$$

- Intuitively, if an e particle and an m particle are adjacent, they can be viewed as a single topological excitation f . This indeed gives the *fusion rule* (algebra):

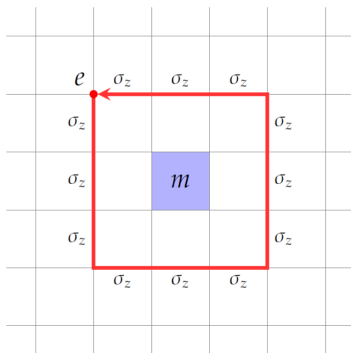
$$e \times m = m \times e = f. \quad (2)$$

Similarly,

$$e \times e = m \times m = f \times f = \mathbb{I}, \quad f \times e = m, \quad f \times m = e. \quad (3)$$

Toric code anyons

- An m particle winding around an e particle creates a phase $i = e^{i\pi/2}$ and vice versa.
- The exchange of two fermions or bosons is restricted by $|\psi_1\psi_2\rangle = \pm |\psi_2\psi_1\rangle$. For anyons it is less-restricted: $|\psi_1\psi_2\rangle = e^{i\theta} |\psi_2\psi_1\rangle$



Fusion channels

- Fusion rules, describing how anyons are brought together, can take the form

$$a \times b = \sum_c N_{ab}^c c, \quad (4)$$

where fusion coefficients $N_{ab}^c = 0, 1, \dots$ describe possible *fusion channels*.

- If for some pair of anyons a and b there are two or more nonzero fusion coefficients, then the model is called *non-Abelian*.
- The anyons in the 2d toric code model is abelian.
- Non-abelian example: $a \times a = 1 + b$, two fusion outcomes.

Braiding anyons

- We seek for a way to encode qubits. For different fusion outcomes, we can define orthonormal states that satisfy

$$\langle ab; c | ab; d \rangle = \delta_{cd}. \quad (5)$$

All these states span *fusion space*.

- The structure of fusion space can be described by F matrix and R matrix. F matrix changes fusion orders of the space $\mathcal{H}_{abc}^d$:

$$|(ab)c; ec; d\rangle = \sum_f (F_{abc}^d)_{ef} |a(bc); af; d\rangle, \quad (6)$$

or in other words, changes basis. A basis is a choice of fusion order, provided that the given anyons and their fusion outcome are fixed.

Braiding anyons

- Diagrammatically, F matrices changes fusion trees.

$$\begin{array}{c} a & b & c \\ & \diagdown & \diagup \\ & e & \\ & | \\ & d \end{array} = \sum_f (F_{abc}^d)_{ef} \begin{array}{c} a & b & c \\ & \diagdown & \diagup \\ & & f \\ & | \\ & d \end{array}$$

- R matrices describe statistics of anyons by

$$|(ba)c; ec; d\rangle = \sum_f R_{ab}^f \delta_{ef} |(ab)c; ec; d\rangle. \quad (7)$$

$$\begin{array}{c} b & a & c \\ & \diagdown & \diagup \\ & \circlearrowleft \\ & e & \\ & | \\ & d \end{array} = R_{ab}^e \begin{array}{c} a & b & c \\ & \diagdown & \diagup \\ & e & \\ & | \\ & d \end{array}$$

Braiding anyons

- To write down the effect of exchanging b and c in the same basis, one first applies F_{abc} to change the basis to $|a(bc); af; d\rangle$, then applies R , and then returns to the original basis with $(F_{abc}^d)^{-1}$,

$$|(ac)b; gb; d\rangle = (F_{acb}^d)^{-1} R F_{abc}^d |(ab)c; ec; d\rangle \quad (8)$$

Pentagon and hexagon identities

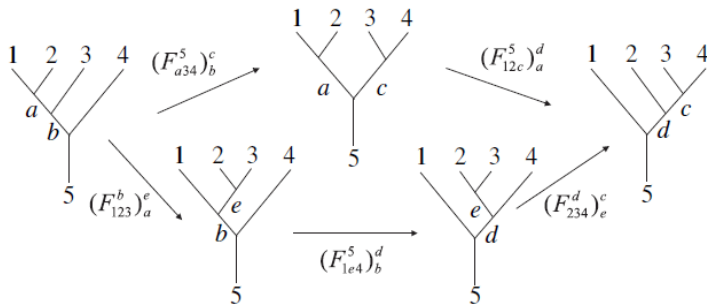


Figure: The diagram of pentagon identity. Similar to idea of bootstrap!

Pentagon and hexagon identities

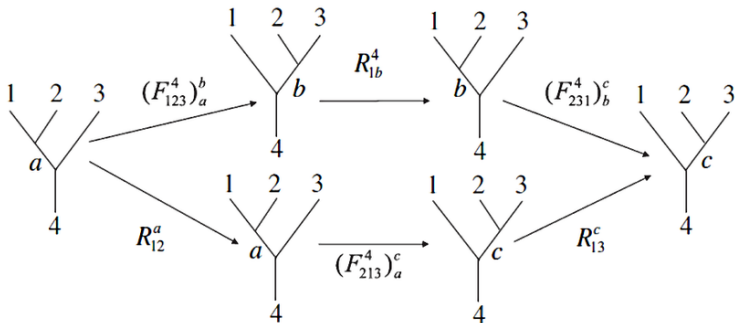


Figure: The diagram of hexagon identity.

Fibonacci anyons

- Fusion rule: $\tau \times \tau = 1 + \tau$. It gives

$$\begin{aligned}\tau \times \tau \times \tau &= 1 + 2\tau \\ \tau \times \tau \times \tau \times \tau &= 2 + 3\tau \\ \tau \times \tau \times \tau \times \tau \times \tau &= 3 + 5\tau\end{aligned}\tag{9}$$

- A basis in this two-dimensional fusion space $\mathcal{H}_{\tau\tau\tau}^\tau$ is given by $|(\tau\tau)\tau; 1\tau; \tau\rangle$ and $|(\tau\tau)\tau; \tau\tau; \tau\rangle$. The other $|\tau(\tau\tau); \tau 1; \tau\rangle, |\tau(\tau\tau); \tau\tau; \tau\rangle$.
- Corresponding F and R matrices are

$$F = F_{\tau\tau\tau}^\tau = \begin{pmatrix} \phi^{-1} & \phi^{-1/2} \\ \phi^{-1/2} & -\phi^{-1} \end{pmatrix}, \quad R = \begin{pmatrix} R_{\tau\tau}^1 & 0 \\ 0 & R_{\tau\tau} \end{pmatrix} = \begin{pmatrix} e^{4\pi i/5} & 0 \\ 0 & e^{-3\pi i/5} \end{pmatrix},\tag{10}$$

where

$$\phi = \frac{1 + \sqrt{5}}{2}$$

is the golden ratio.

Ising anyons

- This model contains three types of anyons, their fusion rules are:

$$\psi \times \psi = 1, \quad \psi \times \sigma = \sigma, \quad \sigma \times \sigma = 1 + \psi \quad (11)$$

- A basis in this constrained fusion space $\mathcal{H}_{\sigma\sigma\sigma}^{\sigma}$ can be

$$|0\rangle_L = |(\sigma\sigma)\sigma; 1\sigma; \sigma\rangle \quad \text{and} \quad |1\rangle_L = |(\sigma\sigma)\sigma; \psi\sigma; \sigma\rangle.$$

The other basis can be

$$|0\rangle_R = |\sigma(\sigma\sigma); \sigma 1; \sigma\rangle \quad \text{and} \quad |1\rangle_R = |\sigma(\sigma\sigma); \sigma\psi; \sigma\rangle.$$

The F matrix and R matrix are given by

$$F = F_{\sigma\sigma\sigma}^{\sigma} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad R_{12} = \begin{pmatrix} R_{\sigma\sigma}^1 & 0 \\ 0 & R_{\sigma\sigma}^{\psi} \end{pmatrix} = e^{-i\pi/8} \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/2} \end{pmatrix}. \quad (12)$$

Ising anyons

$$R_{12}^2 \sim \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad R_{23} = F^{-1} R_{12} F = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad (13)$$

- Explicitly,

$$\begin{aligned} |0\rangle_L &= \frac{1}{\sqrt{2}}(|0\rangle_R + |1\rangle_R), \\ |1\rangle_L &= \frac{1}{\sqrt{2}}(|0\rangle_R - |1\rangle_R) \end{aligned} \quad (14)$$

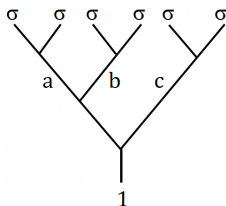
$$R_{12}(|0\rangle_L + |1\rangle_L) = e^{-i\pi/8} |0\rangle_L + e^{3i\pi/8} |1\rangle_L \quad (15)$$

Initialization of a topological quantum computer

- It is convenient to choose the pairwise fusion basis as a computational basis

$$\begin{aligned}
 |00\rangle &= |\sigma\sigma; 1\rangle|\sigma\sigma; 1\rangle|\sigma\sigma; 1\rangle, \\
 |10\rangle &= |\sigma\sigma; \psi\rangle|\sigma\sigma; \psi\rangle|\sigma\sigma; 1\rangle, \\
 |01\rangle &= |\sigma\sigma; 1\rangle|\sigma\sigma; \psi\rangle|\sigma\sigma; \psi\rangle, \\
 |11\rangle &= |\sigma\sigma; \psi\rangle|\sigma\sigma; 1\rangle|\sigma\sigma; \psi\rangle.
 \end{aligned} \tag{16}$$

The total charge is fixed to be 1.



	a	b	c
$ 00\rangle$	1	1	1
$ 10\rangle$	ψ	ψ	1
$ 01\rangle$	1	ψ	ψ
$ 11\rangle$	ψ	1	ψ

Quantum gates – braiding anyons

- In the two-qubit computational basis, the elementary single qubit operations, up to an overall phase, are given by

$$\begin{aligned} X_1 &= R_{23}^2 = F^{-1} R^2 F \otimes \mathbb{I}, & Z_1 &= R_{12}^2 = R^2 \otimes \mathbb{I}, \\ X_2 &= R_{45}^2 = \mathbb{I} \otimes F^{-1} R^2 F, & Z_2 &= R_{56}^2 = \mathbb{I} \otimes R^2. \end{aligned} \tag{17}$$

Quantum gates

- Up to an overall phase, the Hadamard gate on two qubits is given by

$$U_{H,1} = R_{12}R_{23}R_{12} = RF^{-1}RFR \otimes \mathbb{I}, \quad (18)$$

$$U_{H,2} = \mathbb{I} \otimes RF^{-1}RFR. \quad (19)$$

- To implement two-qubit operations, one needs to perform braids outside this group of operations. The natural two-qubit gate to implement with Ising anyons is the controlled phase gate, that is given by the braid

$$U_{CZ} = R_{12}^{-1}R_{34}R_{12}^{-1}. \quad (20)$$

Quantum gates - braiding anyons

$$X_1 = (R_{23})^2$$



$$Z_1 = (R_{12})^2$$



$$U_{H,1} = R_{12}R_{23}R_{12}$$



$$U_{CZ} = R_{12}^{-1}R_{56}^{-1}R_{34}$$



Measurements – fusing anyons

- Computational basis are fusion channel states, so measurement means measuring fusion outcomes.
- For Ising anyons, $\sigma \times \sigma = 1 + \psi$.

1 : nothing left

ψ : neutral fermion excitation

The energy of the final state tells us the fusion channel.

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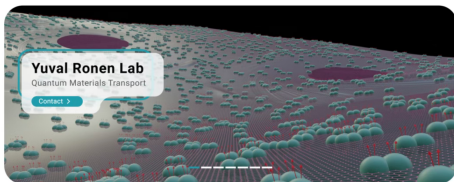
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Summary

- With given anyons and fixed fusion outcomes, different fusion channels can determine a Hilbert space that is the foundation of quantum computation.
- We initialize a topological quantum computer by Ising anyons. Unitary gates can be realized by F matrices and R matrices.

Thanks for your attention!